FEDERICO BARBARROJA

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PROFESSIONAL SUMMARY

AI Engineer | Computer Vision & LLM Solutions

AI Engineer with experience developing and deploying AI solutions in real production environments. Specialized in computer vision and deep learning applied to industrial quality control and the insurance sector, with additional experience in NLP, RAG architectures and LLM-based solutions. Strong background in machine learning, statistics, cloud deployment and software development.

PROFESSIONAL EXPERIENCE

AI Analyst | Inteligencia Analítica – Client: La Segunda Seguros

Rosario, Argentina (Hybrid) | September 2026 - Present

- Training, evaluation and improvement of computer vision models for the automatic detection and classification of vehicle damage, applied to the company's underwriting and claims processes.
- Work on object detection architectures (YOLO) and classification models with ConvNeXt backbones conditioned through FiLM layers, trained on real production images.
- Dataset building and versioning with DVC and Git over Google Cloud Storage, and labeling automation through the Label Studio API.
- Involvement in the migration of ML workflows to OpenShift AI, aimed at reproducible training, deployment and retraining pipelines.

AI Engineer | Terragene

Rosario, Argentina (Hybrid) | May 2026 - July 2026

- Development of AI and computer vision solutions applied to production lines, focused on quality control and automatic defect detection through industrial cameras.
- Training and validation of ML/DL models, dataset building and labeling, and model optimization for production environments.
- Implementation of processing and inference pipelines in Python, integration into industrial systems and model deployment on AWS infrastructure.

Software Developer (Internship) | Renaiss

Rosario, Argentina (Hybrid) | December 2025 - April 2026

- Development and integration of software solutions: structure design, LLM system prompts, OCR integration and consumption of language model APIs.
- Implementation of REST APIs and work with cloud infrastructure and CI/CD pipelines for software, data and AI services.

Data Analyst – Automation & AI | Laboratorios Turner

Rosario, Argentina | June 2024 - January 2026

- Development and implementation of a corporate chatbot based on NLP and a RAG architecture to automate responses to internal and customer queries.
- Design of automated Python pipelines for data extraction, cleaning (ETL) and consolidation, and optimization of inventory and supplier management through SQL Server queries.

EDUCATION

Associate's Degree in Artificial Intelligence | Graduated - July 2026

Universidad Nacional de Rosario (UNR) – Tecnicatura Universitaria en Inteligencia Artificial

TECHNICAL SKILLS

Data & AI: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, NLP, RAG, LangChain, LLM APIs, Vector Databases, MLOps
Computer Vision: OpenCV, CNN, Object Detection, Image Processing, YOLO, SAM, ConvNeXt, Label Studio
Visualization: Matplotlib, Seaborn, Power BI
Programming Languages: Python, SQL, R, Bash
Development & Cloud: Git, DVC, CI/CD, REST APIs, Docker, AWS, GCP (Cloud Storage), OpenShift AI
Tools: SQL Server, Oracle SQL, n8n, Streamlit, Jira, Confluence, Bitbucket
Languages: Spanish (Native), English (B2 - EF SET Certified)